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All-wheel drive system

Abstract

An all-wheel drive system includes a center differential, a limited-slip differential clutch, a front-wheel torque transmission system, a rear-wheel torque transmission system, and a controller. The center differential distributes torque between front and rear wheels of a vehicle. The limited-slip differential clutch limits a differential operation of the center differential in accordance with an engaging pressure, and changes a front-rear torque distribution ratio between the front and rear wheels. The front-wheel torque transmission system transmits torque between the center differential and the front wheels. The rear-wheel torque transmission system transmits torque between the center differential and the rear wheels. The controller adjusts the engaging pressure based on a driving state of the vehicle. Reduction ratios of the front-wheel and rear-wheel torque transmission systems are set different from each other. The center differential is configured such that the front-rear torque distribution ratio is initially unequal and is changeable.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7331896	12/2007	Kroppe	475/221	B60K 17/35
2006/0162981	12/2005	Kurosawa	180/249	B60K 17/3462
2008/0027615	12/2007	Nagura et al.	N/A	N/A
2018/0290540	12/2017	Tsuji	N/A	B60K 23/0808

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
6331826	12/1987	JP	N/A
H02225139	12/1989	JP	N/A
H03118230	12/1990	JP	N/A
H0699758	12/1993	JP	N/A
H0672728	12/1993	JP	N/A
H11-28942	12/1998	JP	N/A
200830626	12/2007	JP	N/A
4519216	12/2009	JP	N/A
2011143855	12/2010	JP	N/A

OTHER PUBLICATIONS

Japanese Office Action for Japanese Patent Application No. 2021-148294, dated May 27, 2025 with English Translation. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority from Japanese Patent Application No. 2021-148294 filed on Sep. 13, 2021, the entire contents of which are hereby incorporated by reference.

BACKGROUND

(2) The disclosure relates to an all-wheel drive system.

(3) An example of an all-wheel drive (AWD) vehicle (or 4-wheel drive (4WD) vehicle) that has been put to use includes drive wheels directly driven by an engine and driven wheels coupled to the engine with a transfer clutch provided therebetween. An engaging force of the transfer clutch is controlled in accordance with, for example, a driving state to adjust a torque allocated to the driven wheels.

(4) Such a coupling all-wheel drive vehicle (torque-split part-time all-wheel drive vehicle) including the transfer clutch is capable of changing a torque distribution ratio between the drive wheels and the driven wheels in the range of 100:0 (drive wheels to driven wheels) to 50:50 by adjusting the engaging pressure (engaging force) of the transfer clutch (by disengaging and engaging the transfer clutch). When, for example, the all-wheel drive vehicle is FF-based, the front-rear torque distribution ratio is changeable in the range of 100:0 (front to rear) and 50:50.

(5) Japanese Unexamined Patent Application Publication (JP-A) No. H11-28942 discloses a four-wheel drive vehicle including a control coupling (transfer clutch) provided on a transmission path along which a driving force for drive wheels is transmitted toward driven wheels to control a transmission ratio of the driving force transmitted to the driven wheels. A gear ratio of the driven wheels is set to a ratio greater than a gear ratio of the drive wheels so that a reduction ratio of the driven wheels is greater than a reduction ratio of the drive wheels, thereby enabling the ratio between the driving force of the driven wheels and the driving force of the drive wheels (front-rear torque distribution ratio) to be changed over a range greater than the range from 0:1 to 1:1.

(6) In one example, this four-wheel drive vehicle is configured such that an output/input gear ratio of a front differential is set to 1 and an output/input gear ratio of a rear differential is set to 1.03 to 1.17. Therefore, the four-wheel drive vehicle is capable of changing the driving state over a wide range from a front-wheel driving state (front biased) to a state close to a rear-wheel driving state (rear biased).

SUMMARY

(7) An aspect of the disclosure provides an all-wheel drive system including a center differential, a limited-slip differential clutch, a front-wheel torque transmission system, a rear-wheel torque transmission system, and a controller. The center differential is configured to distribute torque input thereto between front wheels and rear wheels of a vehicle. The limited-slip differential clutch is configured to limit a differential operation of the center differential in accordance with an engaging pressure and change a front-rear torque distribution ratio between the front wheels and the rear wheels. The front-wheel torque transmission system is configured to transmit torque between the center differential and the front wheels. The rear-wheel torque transmission system is configured to transmit torque between the center differential and the rear wheels. The controller is configured to adjust the engaging pressure of the limited-slip differential clutch based on a driving state of the vehicle. A reduction ratio of the front-wheel torque transmission system and a reduction ratio of the rear-wheel torque transmission system are set to be different from each other. The center differential is configured such that the front-rear torque distribution ratio is initially unequal and is changeable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure

and are incorporated in and constitute a part of this specification. The drawings illustrate an example embodiment and, together with the specification, serve to describe the principles of the disclosure.

(2) FIG. 1 illustrates the overall configuration of an all-wheel drive system according to an embodiment of the disclosure;

(3) FIG. 2 is a flowchart of a process performed by the all-wheel drive system according to the embodiment to suppress (prevent) thermal damage to a limited-slip differential clutch;

(4) FIG. 3 is a graph illustrating the relationship between the input torque, the steering angle, and the clutch engaging force; and

(5) FIG. 4 is a graph illustrating the relationship between the input torque, the steering speed, and the clutch engaging force.

DETAILED DESCRIPTION

(6) According to the all-wheel drive (AWD) vehicle described in JP-A No. H11-28942, the gear ratio of the driven wheels (for example, rear wheels) is set to a ratio greater than the gear ratio of the drive wheels (for example, front wheels) so that the reduction ratio of the driven wheels is greater than the reduction ratio of the drive wheels, thereby enabling the ratio between the torque of the driven wheels and the torque of the drive wheels (front-rear torque distribution ratio) to be changed over a range greater than the range from 0:1 (rear-to-front) to 1:1. In other words, according to the all-wheel drive (AWD) described in JP-A No. H11-28942, the reduction ratio (gear ratio) of the front wheels and the reduction ratio (gear ratio) of the rear wheels are set to different ratios. When, for example, the front wheels are drive wheels, the front-rear torque distribution ratio can be changed from an initial front-biased ratio to a rear-biased ratio by increasing the engaging pressure of the transfer clutch.

(7) However, according to the all-wheel drive vehicle described in JP-A No. H11-28942, the transfer clutch constantly slips because the reduction ratio (gear ratio) of the front wheels and the reduction ratio (gear ratio) of the rear wheels are set to different ratios. Therefore, there are problems of thermal damage, strength, durability, etc. of the transfer clutch. In addition, since a large clutch engaging pressure (engaging force) is generated to achieve a rear-biased state, there is also a problem of loss in, for example, an oil pump that generates a hydraulic pressure.

(8) It is desirable to provide an all-wheel drive system capable of changing a front-rear torque distribution ratio from a front-biased ratio to a rear-biased ratio, suppressing thermal damage to a clutch that changes the front-rear torque distribution ratio, and reducing loss in, for example, an oil pump.

(9) In the following, an embodiment of the disclosure is described in detail with reference to the accompanying drawings. Note that the following description is directed to an illustrative example of the disclosure and not to be construed as limiting to the disclosure. Factors including, without limitation, numerical values, shapes, materials, components, positions of the components, and how the components are coupled to each other are illustrative only and not to be construed as limiting to the disclosure. Further, elements in the following example embodiment which are not recited in a most-generic independent claim of the disclosure are optional and may be provided on an as-needed basis. The drawings are schematic and are not intended to be drawn to scale. Throughout the present specification and the drawings, elements having substantially the same function and configuration are denoted with the same numerals to avoid any redundant description.

(10) The configuration of an all-wheel drive system **1** according to an embodiment will be described with reference to FIG. 1. FIG. 1 illustrates the overall configuration of the all-wheel drive system **1** and an all-wheel drive (AWD) vehicle **3** including the all-wheel drive system **1**.

(11) An engine **20** may be any engine, for example, a horizontally-opposed four-cylinder direct injection gasoline engine. Air introduced into the engine **20** through an air cleaner (not illustrated) is regulated by an electronically controlled throttle valve (hereinafter also referred to simply as a “throttle valve”) **85** provided in an intake pipe, flows through an intake manifold, and is introduced

into each cylinder of the engine **20**. The amount of air introduced through the air cleaner is detected by an air flow meter. The throttle valve **85** is provided with a throttle position sensor **83** that detects the position of the throttle valve **85**. An injector for injecting fuel is attached to each cylinder. A spark plug that ignites an air-fuel mixture and a coil provided with a built-in igniter that applies a high voltage to the spark plug are also attached to each cylinder. The air-fuel mixture containing the introduced air and fuel injected by the injector is ignited by the spark plug and combusted in each cylinder of the engine **20**. Exhaust gas generated by the combustion is discharged through an exhaust pipe.

(12) In addition to the air flow meter and the throttle position sensor **83** described above, a cam angle sensor **81** for cylinder identification of the engine **20** is disposed near a cam shaft of the engine **20**. A crank angle sensor **82** that detects the position of a crank shaft **21** of the engine **20** is disposed near the crank shaft **21**. These sensors are coupled to an engine control unit (hereinafter referred to as an “ECU”) **80** described below. Various other sensors including an accelerator position sensor **84** and a water temperature sensor are also coupled to the ECU **80**. The accelerator position sensor **84** detects the amount by which an accelerator pedal is depressed, that is, an accelerator pedal position. The water temperature sensor detects the temperature of cooling water of the engine **20**.

(13) The crank shaft **21** of the engine **20** is coupled to a continuously variable transmission **30** by which a torque from the engine **20** is converted and output through a torque converter **22** having a clutch function and a torque increasing function and a forward/reverse switching mechanism **27**.

(14) The torque converter **22** mainly includes a pump impeller **23**, a turbine runner **24**, and a stator **25**. The pump impeller **23**, which is coupled to the crank shaft **21**, produces a flow of oil. The turbine runner **24**, which is disposed to face the pump impeller **23**, receives power from the engine **20** through the oil and drives a turbine shaft **31**. The stator **25**, which is positioned between the pump impeller **23** and the turbine runner **24**, regulates an exit (returning) flow from the turbine runner **24** and returns the flow to the pump impeller **23** to produce a torque increasing effect.

(15) The torque converter **22** also includes a lockup clutch **26** that provides a direct input-output coupling. When the lockup clutch **26** is not engaged (in a non-lockup state), the torque converter **22** increases the torque from the engine **20** and transmits the increased torque to the continuously variable transmission **30**. When the lockup clutch **26** is engaged (in a lockup state), the torque converter **22** directly transmits the torque from the engine **20** to the continuously variable transmission **30**. The rotational speed (turbine rotational speed) of the turbine runner **24** included in the torque converter **22** is detected by a turbine rotation sensor **94**. The detected turbine rotational speed is output to a transmission control unit (hereinafter referred to as a “TCU”) **70** described below.

(16) The forward/reverse switching mechanism **27** enables switching between forward and reverse rotations of drive wheels **10**, which are a left front wheel **10FL**, a right front wheel **10FR**, a left rear wheel **10RL**, and a right rear wheel **10RR** (between forward and reverse movements of the vehicle). The forward/reverse switching mechanism **27** mainly includes a double-pinion planetary gear train, a forward clutch **28**, and a reverse brake **29**. The forward/reverse switching mechanism **27** is capable of switching a transmission path along which the engine torque is transmitted by controlling the states of the forward clutch **28** and the reverse brake **29**.

(17) In one example, when a drive (D) range is selected, the forward clutch **28** is engaged and the reverse brake **29** is disengaged so that the rotation of the turbine shaft **31** is directly transmitted to a primary shaft **32** described below to enable the vehicle to move forward. When a reverse (R) range is selected, the forward clutch **28** is disengaged and the reverse brake **29** is engaged so that the planetary gear train is activated and the rotational direction of the primary shaft **32** is reversed to enable the vehicle to move rearward.

(18) When a neutral (N) range or a parking (P) range is selected, the forward clutch **28** and the reverse brake **29** are both disengaged so that the turbine shaft **31** and the primary shaft **32** are

separated from each other (transmission of the engine torque is blocked). Thus, the forward/reverse switching mechanism **27** is set to a neutral state in which no power is transmitted to the primary shaft **32**. The operations of the forward clutch **28** and the reverse brake **29** are controlled by the TCU **70** and a control valve (valve body) **60**.

(19) The continuously variable transmission **30** includes a transmission mechanism (variator) **33** including the primary shaft **32**, which is coupled to the turbine shaft **31** of the torque converter **22** with the forward/reverse switching mechanism **27** provided therebetween, and a secondary shaft **37** disposed parallel to the primary shaft **32**.

(20) The continuously variable transmission **30** includes the primary shaft **32** coupled to the turbine shaft **31** of the torque converter **22** with the forward/reverse switching mechanism **27** provided therebetween and the secondary shaft **37** disposed parallel to the primary shaft **32**. The primary shaft **32** is provided with a primary pulley **34**. The primary pulley **34** includes a fixed sheave **34a** joined to the primary shaft **32** and a movable sheave **34b** facing the fixed sheave **34a** and mounted slidably in an axial direction of the primary shaft **32**. Thus, the primary pulley **34** is configured to have a variable pulley groove width, which is an interval between conical surfaces of the sheaves **34a** and **34b**. The secondary shaft **37** is provided with a secondary pulley **35**. The secondary pulley **35** includes a fixed sheave **35a** joined to the secondary shaft **37** and a movable sheave **35b** facing the fixed sheave **35a** and mounted slidably in an axial direction of the secondary shaft **37**. Thus, the secondary pulley **35** is configured to have a variable pulley groove width.

(21) A chain **36** used to transmit the torque is wrapped around the primary pulley **34** and the secondary pulley **35**. The gear ratio is changed seamlessly by changing the groove widths of the primary pulley **34** and the secondary pulley **35** and changing a ratio between diameters of portions of the chain **36** wrapped around the pulleys **34** and **35** (pulley ratio). When R_p is the diameter of the portion of the chain **36** wrapped around the primary pulley **34** and R_s is the diameter of the portion of the chain **36** wrapped around the secondary pulley **35**, a gear ratio i is determined by $i=R_s/R_p$. Therefore, the gear ratio i is obtained by dividing a primary pulley rotational speed N_p by a secondary pulley rotational speed N_s ($i=N_p/N_s$).

(22) The primary pulley **34** (movable sheave **34b**) has a hydraulic chamber **34c**. The secondary pulley **35** (movable sheave **35b**) has a hydraulic chamber **35c**. The groove widths of the primary pulley **34** and the secondary pulley **35** are set and changed by adjusting a primary hydraulic pressure applied to the hydraulic chamber **34c** of the primary pulley **34** and a secondary hydraulic pressure applied to the hydraulic chamber **35c** of the secondary pulley **35**.

(23) The secondary shaft **37** of the continuously variable transmission **30** is coupled to a center differential **40** (described in detail below) that distributes the input torque between the front wheels and the rear wheels. The torque converted by the continuously variable transmission **30** is distributed between the front wheels and the rear wheels at a predetermined torque distribution ratio (described in detail below) by the center differential **40**. The torque allocated to the front wheels by the center differential **40** is transmitted to a front differential **44** through a transfer gearset **401** (transfer drive gear **401a** and transfer driven gear **401b**) and a front drive shaft **43**. The front differential **44** is, for example, a bevel-gear differential. The torque from the front differential **44** is transmitted to the left front wheel **10FL** through a left-front-wheel drive shaft **45L** and to the right front wheel **10FR** through a right-front-wheel drive shaft **45R**. In one embodiment, the transfer gearset **401** (transfer drive gear **401a** and transfer driven gear **401b**), the front drive shaft **43**, the front differential **44**, the left-front-wheel drive shaft **45L**, and the right-front-wheel drive shaft **45R** may serve as a “front-wheel torque transmission system”.

(24) The torque allocated to the rear wheels by the center differential **40** is transmitted to a rear differential **47** through a propeller shaft **46** that extends toward the rear of the vehicle. The rear differential **47** is coupled to a left-rear-wheel drive shaft **48L** and a right-rear-wheel drive shaft **48R**. The torque from the rear differential **47** is transmitted to the left rear wheel **10RL** through the left-rear-wheel drive shaft **48L** and to the right rear wheel **10RR** through the right-rear-wheel drive

shaft **48R**. In one embodiment, the propeller shaft **46**, the rear differential **47**, the left-rear-wheel drive shaft **48L**, and the right-rear-wheel drive shaft **48R** may serve as a “rear-wheel torque transmission system”.

(25) The reduction ratio (total gear ratio) of the front-wheel torque transmission system and the reduction ratio (total gear ratio) of the rear-wheel torque transmission system are set to different ratios. For example, in the present embodiment, the gear ratios of the transfer gearset **401**, the front differential **44**, and the rear differential **47** are set so that the reduction ratio (total gear ratio) of the front-wheel torque transmission system is less than the reduction ratio (total gear ratio) of the rear-wheel torque transmission system. In other words, in the present embodiment, the gear ratios are set so that the following expression is satisfied: (gear ratio of the transfer gearset **401**) $\times$ (gear ratio of the front differential **44**) $<$ (gear ratio of the rear differential **47**). Therefore, the following expression is satisfied: (rotational speeds of the rear wheels **10RL** and **10RR**) $<$ (rotational speeds of the front wheels **10FL** and **10FR**).

(26) A secondary shaft **37** is rotatably inserted in the center differential **40** from the front of a carrier **402** that is rotatably housed in the center differential **40**. The propeller shaft **46** is rotatably inserted in the center differential **40** from the rear.

(27) A first sun gear **403** having a small diameter is attached to the rear end of the secondary shaft **37** at the input side, and a second sun gear **404** having a large diameter is attached to the front end of the propeller shaft **46** through which torque is output to the rear wheels **10RL** and **10RR**. The first sun gear **403** and the second sun gear **404** are disposed in the carrier **402**.

(28) The first sun gear **403** meshes with first pinion gears **405** having a large diameter to form a first gear train, and the second sun gear **404** meshes with second pinion gears **406** having a small diameter to form a second gear train. The first pinion gears **405** and the second pinion gears **406** are permanently affixed to each other so that a plurality of pairs (for example, three pairs) of pinion gears are rotatably supported by the carrier **402**. The transfer drive gear **401a** is coupled to the front end of the carrier **402**, so that torque is output from the carrier **402** to the front wheels.

(29) Thus, the center differential **40** has a compound planetary gear system having no ring gear and configured such that the torque from the secondary shaft **37** is transmitted to the first sun gear **403** and output to the propeller shaft **46** from the second sun gear **404**, and is also output to the front drive shaft **43** from the carrier **402** through the transfer drive gear **401a** and the transfer driven gear **401b**.

(30) The center differential **40** having the compound planetary gear system provides a differential function by appropriately setting the numbers of teeth of the first and second sun gears **403** and **404** and a plurality of pairs of first and second pinion gears **405** and **406** disposed around the sun gears **403** and **404**.

(31) By appropriately setting the meshing pitch circle radii of the first and second sun gears **403** and **404** and the first and second pinion gears **405** and **406** (=torque ratio), a reference (initial) torque distribution may be set to an equal torque distribution of 50:50 (front to rear) or an unequal torque distribution that is front biased or rear biased. In the present embodiment, the front-rear torque distribution ratio of the center differential **40** in an initial state is set to for example, 80:20 (front to rear) and is front biased (torque allocated to the front wheels $>$ torque allocated to the rear wheels).

(32) In one example, the center differential **40** is configured such that the front-rear torque distribution ratio is set to an unequal ratio in the initial state and is changeable (that is, changeable by changing the meshing pitch circle radii of the first and second sun gears **403** and **404** and the first and second pinion gears **405** and **406**).

(33) A limited-slip differential (LSD) clutch **407** that employs a hydraulic multiple disc clutch is provided between the carrier **402** of the center differential **40** and the propeller shaft **46**. The limited-slip differential clutch **407** limits the differential operation of the center differential **40** and changes the front-rear torque distribution ratio between the front wheels and the rear wheels (torque

distribution between the front and rear wheels). The front-rear torque distribution ratio is changeable in the range of, for example, 80:20 to 20:80 by controlling an engaging force of the limited-slip differential clutch **407**. An engaging pressure (engaging force) of the limited-slip differential clutch **407** is controlled by the TCU **70** and the control valve (valve body) **60**.

(34) According to a power train of a torque transmission system configured as described above, when a selection lever is operated to select the D range, for example, the engine torque is input to the primary shaft **32** of the continuously variable transmission **30**. The torque converted by the continuously variable transmission **30** is output from the secondary shaft **37** and distributed between the front wheels and the rear wheels by the center differential **40**. When a clutch torque is applied to the limited-slip differential clutch **407**, torque is distributed between the front drive shaft **43** and the propeller shaft **46** in accordance with the applied clutch torque.

(35) Torque allocated to the front wheels is transmitted to the front drive shaft **43** and distributed between the left and right front wheels **10FL** and **10FR** by the front differential **44**. Torque allocated to the rear wheels is transmitted to the propeller shaft **46** and distributed between the rear wheels **10RL** and **10RR** by the rear differential **47**.

(36) The wheels **10FL**, **10FR**, **10RL**, and **10RR** (hereinafter also referred to collectively as wheels **10**) are respectively equipped with brakes **11FL**, **11FR**, **11RL**, and **11RR** (hereinafter also referred to collectively as brakes **11**) for braking the wheels **10FL**, **10FR**, **10RL**, and **10RR**. The wheels **10FL**, **10FR**, **10RL**, and **10RR** are also respectively equipped with wheel speed sensors **12FL**, **12FR**, **12RL**, and **12RR** (hereinafter also referred to collectively as wheel speed sensors **12**) that detect rotational speeds of the wheels **10FL**, **10FR**, **10RL**, and **10RR**.

(37) In the present embodiment, disc brakes are used as the brakes **11**. Each brake **11** includes a brake disc attached to the corresponding wheel **10** of the AWD vehicle **3** and a brake caliper in which a brake pad and a wheel cylinder are mounted. During braking, the brake pad is hydraulically pressed against the brake disc, and the resultant frictional force brakes the wheel **10** coupled to the brake disc. Although a disc brake is used as each brake **11** in the present embodiment, other brakes, such as a drum brake that presses a friction material against an inner peripheral surface of a drum, may instead be used.

(38) Each wheel speed sensor **12** is a contactless sensor that detects a change in a magnetic field produced by a rotor (gear rotor or magnetic rotor) that rotates together with the corresponding wheel **10**. For example, a semiconductor sensor that detects the rotation of the rotor with a Hall element or an MR element may be suitably used.

(39) The AWD vehicle **3** also includes a vehicle dynamics control (VDC) unit (hereinafter abbreviated as "VDCU") **50** that prevents skidding and ensures high traveling stability when, for example, the AWD vehicle **3** enters a corner at an excessive speed or when the position (behavior) of the AWD vehicle **3** becomes unstable due to an abrupt steering operation.

(40) The hydraulic pressures allowing the continuously variable transmission **30** to change the gear ratio, that is, the primary hydraulic pressure and the secondary hydraulic pressure described above, are controlled by the control valve (valve body) **60**. The valve body **60** adjusts the hydraulic pressure supplied from an oil pump **62** by opening and closing an oil passage formed in the valve body **60** by using a spool valve and a solenoid valve (electromagnetic valve) that moves the spool valve, and supplies the adjusted hydraulic pressure to each of the hydraulic chamber **34c** of the primary pulley **34** and the hydraulic chamber **35c** of the secondary pulley **35**. Similarly, the control valve **60** adjusts the hydraulic pressure supplied from the oil pump **62** by opening and closing an oil passage formed in the valve body **60** by using a spool valve and a solenoid valve (electromagnetic valve) that moves the spool valve, and supplies the adjusted hydraulic pressure as a hydraulic pressure for adjusting the engaging pressure (engaging force) of the limited-slip differential clutch **407**. The solenoid valve used to adjust the hydraulic pressure supplied to the limited-slip differential clutch **407** may be, for example, a duty solenoid capable of controlling the amount by which the solenoid is driven in accordance with a duty ratio of a voltage applied thereto.

(41) Transmission control of the continuously variable transmission **30** is performed by the TCU **70**. In one example, the TCU **70** adjust the hydraulic pressure supplied to each of the hydraulic chamber **34c** of the primary pulley **34** and the hydraulic chamber **35c** of the secondary pulley **35** by controlling the operation of the solenoid valve (electromagnetic valve) included in the above-described valve body **60**, and thereby changes the gear ratio of the continuously variable transmission **30**. Similarly, the TCU **70** adjusts the hydraulic pressure supplied to the limited-slip differential clutch **407** by controlling the operation of the solenoid valve included in the above-described control valve **60**, and thereby adjusts the distribution ratio between the torque transmitted to the front wheels **10FL** and **10FR** and the torque transmitted to the rear wheels **10RL** and **10RR** (front-rear torque distribution ratio).

(42) A shift lever (selection lever) **75** that receives an operation performed by the driver to select an operational state (range) of the continuously variable transmission **30** is provided on, for example, a floor (center console) of the vehicle. A range switch **93** that moves together with the shift lever **75** and detects the selected position of the shift lever **75** is coupled to the shift lever **75**. The range switch **93** is also coupled to the TCU **70**, and the detected selected position of the shift lever **75** is read by the TCU **70**. The shift lever **75** can be selectively switched between the drive (D) range, a manual (M) range, the parking (P) range, the reverse (R) range, and the neutral (N) range. A selecting switch mechanism may be used instead of the shift lever **75**.

(43) When the shift lever **75** is operated to select the D range (forward driving range), oil is supplied to the hydraulic chamber of the forward clutch **28** and discharged from the hydraulic chamber of the reverse brake **29**. Accordingly, the forward clutch **28** is engaged and the reverse brake **29** is disengaged to enable forward driving of the vehicle. When the shift lever **75** is operated to select the R range (reverse driving range), oil is supplied to the hydraulic chamber of the reverse brake **29** and discharged from the hydraulic chamber of the forward clutch **28**. Accordingly, the reverse brake **29** is engaged and the forward clutch **28** is disengaged to enable reverse driving of the vehicle. When the shift lever **75** is operated to select the N range or the P range, oil is discharged from both the hydraulic chamber of the forward clutch **28** and the hydraulic chamber of the reverse brake **29**. Accordingly, the forward clutch **28** and the reverse brake **29** are both disengaged (transmission of the engine torque is blocked) so that the continuously variable transmission **30** is set to a neutral state.

(44) As described above, the TCU **70** performs the transmission control of the continuously variable transmission **30**, the engagement/disengagement control (forward-reverse switch control) of the forward clutch **28** and the reverse brake **29**, and the engaging pressure control (front-rear torque distribution control) of the limited-slip differential clutch **407**. The TCU **70** is coupled to the ECU **80** that controls the overall operation of the engine **20**, the VDCU **50**, and other units through, for example, a controller area network (CAN) **100** that enables intercommunication therebetween.

(45) Each of the TCU **70**, the ECU **80**, and the VDCU **50** includes a microprocessor that performs computation, an electrically erasable programmable read-only memory (EEPROM) that stores a program for causing the microprocessor to execute each process, a random access memory (RAM) that stores various data, such as computation results, a backup RAM that holds contents stored therein by using a battery, and an input/output interface (I/F).

(46) The ECU **80** performs cylinder identification based on the output of the cam angle sensor **81** and determines the engine rotational speed based on a change in the rotational position of the crank shaft detected from the output of the crank angle sensor **82**. The ECU **80** also acquires various information including an amount of air introduced, the accelerator pedal position, the air-fuel ratio of the air-fuel mixture, and a water temperature based on detection signals input from various sensors described above. The ECU **80** controls the overall operation of the engine **20** by controlling an amount of fuel injection, ignition timing, and various devices including the throttle valve **85** based on the acquired information.

(47) The ECU **80** also transmits various information including the engine water temperature

(cooling water temperature), the accelerator pedal position, the engine rotational speed, and the engine torque to the TCU **70** through the CAN **100**.

(48) The VDCU **50** is coupled to the four wheel speed sensors **12FL**, **12FR**, **12RL**, and **12RR**, a steering angle sensor **16**, an acceleration (G) sensor **55**, a pitch angle sensor **56**, and a brake switch **57**. As described above, the wheel speed sensors **12FL**, **12FR**, **12RL**, and **12RR** respectively detect the rotational states of the wheels **10FL**, **10FR**, **10RL**, and **10RR** by detecting the rotations of gears attached to the centers of the wheels **10FL**, **10FR**, **10RL**, and **10RR** with, for example, magnetic pickups. The acceleration sensor **55** detects accelerations applied to the AWD vehicle **3** in the front-rear and sideways directions. The pitch angle sensor **56** detects a rotation angle around an axis passing through the center of gravity of the AWD vehicle **3** and extending through the vehicle body in the vehicle width direction (road gradient). The steering angle sensor **16** detects a steered angle of the front wheels **10FL** and **10FR** that are steered wheels (that is, a steering angle of a steering wheel **15**) by detecting a rotation angle of a pinion shaft.

(49) The VDCU **50** brakes the vehicle by driving brake actuators in accordance with an amount by which a brake pedal is operated (depressed). The VDCU **50** also detects the behavior of the vehicle by using various sensors (for example, the wheel speed sensors **12**, the steering angle sensor **16**, the acceleration sensor **55**, the pitch angle sensor **56**, and a yaw rate sensor) and performs brake control based on automatic pressurization and torque control of the engine **20** to suppress skidding and ensure high vehicle stability during cornering. In other words, the VDCU **50** prevents skidding and ensures high traveling stability when, for example, the vehicle enters a corner at an excessive speed or when the position (behavior) of the vehicle becomes unstable due to an abrupt steering operation. In one example, the VDCU **50** detects the position (behavior) of the vehicle by using the above-described sensors, for example, and performs an automatic control operation depending on a driving state. For example, when oversteering is detected, the VDCU **50** brakes one of the front wheels **10FL** and **10FR** that is on the outer side of the corner. When understeering is detected, the VDCU **50** reduces the engine power and brakes one of the rear wheels **10RL** and **10RR** that is on the inner side of the corner. The VDCU **50** also has a function of an antilock braking system (ABS) and a function of a traction control system (TCS) in addition to the VDC (skidding prevention) function.

(50) The VDCU **50** transmits various information including the rotational speeds of the wheels **10**, the steering angle, the accelerations, and the pitch angle (road gradient) that have been detected and braking information to the TCU **70** through the CAN **100**.

(51) The TCU **70** is coupled not only to the above-described turbine rotation sensor **94** but also to an oil temperature sensor **91** that detects an oil temperature of the continuously variable transmission **30** (that is, an oil temperature of the limited-slip differential clutch **407**); an output shaft rotation sensor **92** that detects a rotational speed of the secondary shaft **37** (rotational speed of the secondary pulley **35**); and a range switch **93** that detects the selected position of the shift lever.

(52) As above-described, the TCU **70** receives various information including the rotational speeds of the wheels **10**, the steering angle, the accelerations, the pitch angle (road gradient), and the braking information from the VDCU **50** through the CAN **100**, and also receives information including the engine water temperature, the accelerator pedal position, the engine rotational speed, and the engine torque from the ECU **80** through the CAN **100**.

(53) The TCU **70** automatically seamlessly changes the gear ratio depending on the driving state (for example, the accelerator position and the vehicle speed) of the AWD vehicle **3** in accordance with a shift map. The shift map is stored in, for example, the EEPROM of the TCU **70**.

(54) The TCU **70** performs limited-slip differential clutch control (front-rear torque distribution control) based on various information acquired from the above-described sensors. For example, the TCU **70** adjusts the engaging force of the limited-slip differential clutch **407** (that is, the front-rear torque distribution ratio) based on the driving state of the AWD vehicle **3** (for example, the steering angle and the pitch angle). In one embodiment, the TCU **70** may serve as a “controller”.

(55) In particular, the TCU **70** is capable of changing the front-rear torque distribution ratio from a front-biased ratio to a rear-biased ratio and has functions of suppressing thermal damage to the limited-slip differential clutch **407** that changes the front-rear torque distribution ratio and reducing loss in, for example, the oil pump **62**. The TCU **70** realizes these functions by causing the microprocessor to execute a program stored in, for example, the EEPROM.

(56) For example, during accelerated cornering or accelerated climbing, the TCU **70** adjusts the engaging pressure (engaging force) of the limited-slip differential clutch **407** to change the front-rear torque distribution ratio to a rear-biased ratio. In a driving state other than accelerated cornering or accelerated climbing, the TCU **70** disengages the limited-slip differential clutch **407**. Accordingly, the front-rear torque distribution ratio is set to a front-biased (initial) ratio.

(57) In one example, the TCU **70** sets the front-rear torque distribution ratio to a rear-biased ratio by increasing the engaging force (engaging pressure) of the limited-slip differential clutch **407** when the torque input to the center differential **40** (engine torque $\times$ gear ratio) is greater than or equal to a predetermined torque (for example, 10 Nm) and the steering angle of the steering wheel **15** is greater than or equal to a predetermined angle (for example, 1°), or when the torque input to the center differential **40** is greater than or equal to a predetermined torque (for example, 10 Nm) and the pitch angle of the AWD vehicle **3** (road gradient) is greater than or equal to a predetermined angle (for example, 1°).

(58) The EEPROM of the TCU **70**, for example, stores a map (target clutch torque map) defining the relationship between the torque input to the center differential **40**, the steering angle of the steering wheel **15**, the pitch angle of the AWD vehicle **3** (gradient), and the target clutch torque of the limited-slip differential clutch **407**. The target clutch torque of the limited-slip differential clutch **407** is determined based on the steering angle and the pitch angle by referring to the target clutch torque map.

(59) As illustrated in FIG. **3**, the TCU **70** increases the engaging pressure (engaging force) of the limited-slip differential clutch **407** in proportion to the steering angle as the steering angle increases. However, when a large clutch engaging force is applied during low-speed cornering, it becomes difficult to turn smoothly (tight corner braking phenomenon). Therefore, the clutch engaging force may not be increased when the steering speed is low. Accordingly, as illustrated in FIG. **4**, the TCU **70** increases the engaging pressure (engaging force) of the limited-slip differential clutch **407** in proportion to the steering angular speed as the steering angular speed increases. FIG. **3** is a graph illustrating the relationship between the input torque, the steering angle, and the clutch engaging force. FIG. **4** is a graph illustrating the relationship between the input torque, the steering speed (steering angular speed), and the clutch engaging force.

(60) The TCU **70** disengages the limited-slip differential clutch **407** when the oil temperature of the limited-slip differential clutch **407** (continuously variable transmission **30**) is higher than or equal to a predetermined temperature (for example, 80° C.)

(61) The operation of the all-wheel drive system **1** will now be described with reference to FIG. **2**. FIG. **2** is a flowchart of a process performed by the all-wheel drive system **1** to suppress (prevent) thermal damage to the limited-slip differential clutch **407**. This process is repeatedly performed mainly by the TCU **70** at a predetermined timing.

(62) In step **S100**, it is determined whether the oil temperature of the limited-slip differential clutch **407** (continuously variable transmission **30**) is higher than or equal to a predetermined temperature (for example, 80° C.) When the oil temperature of the limited-slip differential clutch **407** is higher than or equal to the predetermined temperature, a target clutch torque of the limited-slip differential clutch **407** is set to zero in step **S110**. After that, in step **S112**, a target hydraulic pressure of the limited-slip differential clutch **407** is set to zero, and the limited-slip differential clutch **407** is disengaged. After that, the process is temporarily stopped. When the oil temperature of the limited-slip differential clutch **407** is less than the predetermined temperature, the process proceeds to step **S102**.

(63) In step **S102**, it is determined whether the torque input to the center differential **40** is greater than or equal to a predetermined torque (for example, 10 Nm). When the torque input to the center differential **40** is less than the predetermined torque, the target clutch torque of the limited-slip differential clutch **407** is set to zero in step **S110**. After that, in step **S112**, the target hydraulic pressure of the limited-slip differential clutch **407** is set to zero, and the limited-slip differential clutch **407** is disengaged. After that, the process is temporarily stopped. When the torque input to the center differential **40** is greater than or equal to the predetermined torque, the process proceeds to step **S104**.

(64) In step **S104**, it is determined whether the steering angle of the steering wheel **15** is greater than or equal to a predetermined angle (for example, 1°). When the steering angle of the steering wheel **15** is less than the predetermined angle, the process proceeds to step **S106**. When the steering angle of the steering wheel **15** is greater than or equal to the predetermined angle, the process proceeds to step **S108**.

(65) In step **S106**, it is determined whether the pitch angle of the AWD vehicle **3** (road gradient) is greater than or equal to a predetermined angle (for example, 1°). When the pitch angle of the AWD vehicle **3** is less than the predetermined angle, the target clutch torque of the limited-slip differential clutch **407** is set to zero in step **S110**. After that, in step **S112**, the target hydraulic pressure of the limited-slip differential clutch **407** is set to zero, and the limited-slip differential clutch **407** is disengaged. After that, the process is temporarily stopped. When the pitch angle of the AWD vehicle **3** is greater than or equal to the predetermined angle, the process proceeds to step **S108**.

(66) In step **S108**, the target clutch torque of the limited-slip differential clutch **407** is determined by referring to the map (target clutch torque map) defining the relationship between the torque input to the center differential **40**, the steering angle of the steering wheel **15**, the pitch angle of the AWD vehicle **3** (road gradient), and the target clutch torque of the limited-slip differential clutch **407**.

(67) Subsequently, in step **S112**, a target clutch hydraulic pressure for the limited-slip differential clutch **407** is set in accordance with the target clutch torque of the limited-slip differential clutch **407** determined in step **S108**. Then, the hydraulic pressure (that is, the engaging pressure) of the limited-slip differential clutch **407** is controlled so that the actual hydraulic pressure becomes equal to the target clutch hydraulic pressure.

(68) As described in detail above, according to the present embodiment, the reduction ratio of the front-wheel torque transmission system and the reduction ratio of the rear-wheel torque transmission system are set to different ratios, and the front-rear torque distribution ratio of the center differential **40** is initially unequal and is changeable. Since the front-rear torque distribution ratio is initially unequal and is changeable, a variable range of the front-rear torque distribution ratio may be reduced by increasing the initial torque allocated to the driven wheels (for example, the rear wheels) as compared to that in a coupling all-wheel drive vehicle in which the front and rear reduction ratios (gear ratios) are set to different ratios. Accordingly, the engaging pressure (engaging force) of the limited-slip differential clutch **407** can be reduced. Therefore, the amount of heat generated in the limited-slip differential clutch **407** can be reduced and thermal damage to the limited-slip differential clutch **407** can be suppressed (reduced). In addition, since the engaging pressure (engaging force) of the limited-slip differential clutch **407** can be reduced, loss in, for example, the oil pump **62** that generates (increases) the hydraulic pressure can be reduced. As a result, the front-rear torque distribution ratio can be changed from a front-biased ratio to a rear-biased ratio, thermal damage to the limited-slip differential clutch **407** that changes the front-rear torque distribution ratio can be suppressed, and loss in, for example, the oil pump **62** can be reduced.

(69) According to the present embodiment, the front-rear torque distribution ratio of the center differential **40** is initially front biased (torque allocated to the front wheels > torque allocated to the

rear wheels). In addition, the gear ratios of the transfer gaset **401**, the front differential **44**, and/or the rear differential **47** are set so that the reduction ratio of the front-wheel torque transmission system is less than the reduction ratio of the rear-wheel torque transmission system. Therefore, by increasing the engaging pressure of the limited-slip differential clutch **407**, the front-rear torque distribution ratio can be changed from a front-biased ratio (initial ratio) to a rear-biased ratio.

(70) According to the present embodiment, the engaging pressure (engaging force) of the limited-slip differential clutch **407** is adjusted during accelerated cornering or accelerated climbing, and the limited-slip differential clutch **407** is disengaged in a driving state other than accelerated cornering or accelerated climbing. Therefore, the region in which the limited-slip differential clutch **407** is activated can be limited, and thermal damage to the limited-slip differential clutch **407** can be more effectively suppressed (reduced).

(71) According to the present embodiment, the engaging pressure (engaging force) of the limited-slip differential clutch **407** is increased when the torque input to the center differential **40** is greater than or equal to a predetermined torque and the steering angle of the steering wheel **15** is greater than or equal to a predetermined angle, or when the torque input to the center differential **40** is greater than or equal to the predetermined torque and the pitch angle of the AWD vehicle **3** (gradient) is greater than or equal to a predetermined angle. Thus, a suitable front-rear torque distribution ratio can be set in accordance with the driving state of the AWD vehicle **3** (as necessary).

(72) According to the present embodiment, the limited-slip differential clutch **407** is disengaged when the oil temperature of the limited-slip differential clutch **407** (continuously variable transmission **30**) is higher than or equal to a predetermined temperature. Therefore, thermal damage to the limited-slip differential clutch **407** may be prevented.

(73) Although an embodiment of the disclosure has been described, the disclosure is not limited to the above-described embodiment, and various modifications are possible. For example, in the above-described embodiment, the front-rear torque distribution ratio of the center differential **40** is set to 80:20 (front biased) in the initial state. However, the initial front-rear torque distribution ratio of the center differential **40** is not limited to that in the above-described embodiment (80:20), and may be set to any ratio depending on, for example, conditions. In addition, according to the above-described embodiment, the front-rear torque distribution ratio is changeable in the range of 80:20 to 20:80 by controlling the engaging force of the limited-slip differential clutch **407**. However, the variable range of the front-rear torque distribution ratio is not limited to that in the above-described embodiment (80:20 to 20:80) and may be set to any range in accordance with, for example, conditions.

(74) In the above-described embodiment, the reduction ratio (total gear ratio) of the front-wheel torque transmission system is set to a ratio less than the reduction ratio (total gear ratio) of the rear-wheel torque transmission system. However, the reduction ratio (total gear ratio) of the front-wheel torque transmission system may instead be set to a ratio greater than the reduction ratio (total gear ratio) of the rear-wheel torque transmission system.

(75) In the above-described embodiment, the engaging pressure (engaging force) of the limited-slip differential clutch **407** is adjusted in accordance with the torque input to the center differential **40**, the steering angle of the steering wheel **15**, the pitch angle of the AWD vehicle **3**, and the oil temperature of the limited-slip differential clutch **407** (continuously variable transmission **30**). However, the operational conditions under which the limited-slip differential clutch **407** is adjusted are not limited to those in the above-described embodiment.

(76) In the above-described embodiment, a chain continuously variable transmission (CVT) is described as an example of an automatic transmission. However, a belt continuously variable transmission or a toroidal continuously variable transmission, for example, may be used instead of the chain continuously variable transmission. Alternatively, a stepped automatic transmission (AT), a manual transmission (MT), or a dual clutch transmission (DCT) may be used instead of the

continuously variable transmission.

(77) The above-described configurations of the torque transmission systems (for example, configurations of gear trains and shafts) are examples, and are not limited to those in the above-described embodiment. The configuration of the control system is also not limited to that in the above-described embodiment (that is, the configuration in which the control units are coupled to each other by the CAN **100**).

(78) The TCU **70** illustrated in FIG. **1** can be implemented by circuitry including at least one semiconductor integrated circuit such as at least one processor (e.g., a central processing unit (CPU)), at least one application specific integrated circuit (ASIC), and/or at least one field programmable gate array (FPGA). At least one processor can be configured, by reading instructions from at least one machine readable tangible medium, to perform all or a part of functions of the TCU **70**. Such a medium may take many forms, including, but not limited to, any type of magnetic medium such as a hard disk, any type of optical medium such as a CD and a DVD, any type of semiconductor memory (i.e., semiconductor circuit) such as a volatile memory and a non-volatile memory. The volatile memory may include a DRAM and a SRAM, and the non-volatile memory may include a ROM and a NVRAM. The ASIC is an integrated circuit (IC) customized to perform, and the FPGA is an integrated circuit designed to be configured after manufacturing in order to perform all or a part of the functions of the modules illustrated in FIG. **1**.

Claims

1. An all-wheel drive system comprising: a center differential configured to distribute torque input thereto between front wheels and rear wheels of a vehicle; a limited-slip differential clutch configured to limit a differential operation of the center differential in accordance with an engaging pressure and change a front-rear torque distribution ratio between the front wheels and the rear wheels; a front-wheel torque transmission system configured to transmit torque between the center differential and the front wheels; a rear-wheel torque transmission system configured to transmit torque between the center differential and the rear wheels; and a controller configured to adjust the engaging pressure of the limited-slip differential clutch based on a driving state of the vehicle, wherein a reduction ratio of the front-wheel torque transmission system and a reduction ratio of the rear-wheel torque transmission system are set to be different from each other, and wherein the center differential is configured such that the front-rear torque distribution ratio is initially unequal and is changeable.

2. The all-wheel drive system according to claim 1, wherein the front-rear torque distribution ratio of the center differential is initially front biased, and wherein a gear ratio of a transfer gearset, a gear ratio of a front differential, and/or a gear ratio of a rear differential are set so that the reduction ratio of the front-wheel torque transmission system is less than the reduction ratio of the rear-wheel torque transmission system.

3. The all-wheel drive system according to claim 2, wherein the controller adjusts the engaging pressure of the limited-slip differential clutch when the driving state of the vehicle is accelerated cornering or accelerated climbing, and disengages the limited-slip differential clutch in a case where the driving state of the vehicle is other than the accelerated cornering or the accelerated climbing.

4. The all-wheel drive system according to claim 3, wherein the controller disengages the limited-slip differential clutch in a case where an oil temperature of the limited-slip differential clutch is higher than or equal to a predetermined temperature.

5. The all-wheel drive system according to claim 3, wherein the controller increases the engaging pressure of the limited-slip differential clutch in a case where the torque input to the center differential is greater than or equal to a predetermined torque and a steering angle of a steering wheel is greater than or equal to a predetermined angle, or in a case where the torque input to the

center differential is greater than or equal to a predetermined torque and a pitch angle of the vehicle is greater than or equal to a predetermined angle.

6. The all-wheel drive system according to claim 5, wherein the controller disengages the limited-slip differential clutch in a case where an oil temperature of the limited-slip differential clutch is higher than or equal to a predetermined temperature.

7. The all-wheel drive system according to claim 1, wherein the controller adjusts the engaging pressure of the limited-slip differential clutch in a case where the driving state of the vehicle is accelerated cornering or accelerated climbing, and disengages the limited-slip differential clutch in a case where the driving state of the vehicle is other than the accelerated cornering or the accelerated climbing.

8. The all-wheel drive system according to claim 7, wherein the controller increases the engaging pressure of the limited-slip differential clutch in a case where the torque input to the center differential is greater than or equal to a predetermined torque and a steering angle of a steering wheel is greater than or equal to a predetermined angle, or in a case where the torque input to the center differential is greater than or equal to a predetermined torque and a pitch angle of the vehicle is greater than or equal to a predetermined angle.

9. The all-wheel drive system according to claim 8, wherein the controller disengages the limited-slip differential clutch in a case where an oil temperature of the limited-slip differential clutch is higher than or equal to a predetermined temperature.

10. The vehicle comprising the all-wheel drive system according to claim 1.

11. The all-wheel drive system according to claim 1, wherein the controller is configured to change the front-rear torque distribution ratio from a front-biased ratio to a rear-biased ratio.

12. An all-wheel drive system comprising: a center differential configured to distribute torque input thereto between front wheels and rear wheels of a vehicle; a limited-slip differential clutch configured to limit a differential operation of the center differential in accordance with an engaging pressure and change a front-rear torque distribution ratio between the front wheels and the rear wheels; a front-wheel torque transmission system configured to transmit torque between the center differential and the front wheels; a rear-wheel torque transmission system configured to transmit torque between the center differential and the rear wheels; and a controller configured to adjust the engaging pressure of the limited-slip differential clutch based on a driving state of the vehicle, wherein a reduction ratio of the front-wheel torque transmission system and a reduction ratio of the rear-wheel torque transmission system are set to be different from each other, and wherein the center differential is configured such that the front-rear torque distribution ratio is initially unequal and is changeable, wherein the controller adjusts the engaging pressure of the limited-slip differential clutch in a case where the driving state of the vehicle is accelerated cornering or accelerated climbing, and disengages the limited-slip differential clutch in a case where the driving state of the vehicle is other than the accelerated cornering or the accelerated climbing, wherein the controller disengages the limited-slip differential clutch in a case where an oil temperature of the limited-slip differential clutch is higher than or equal to a predetermined temperature.

13. A program for an all-wheel drive system stored in a non-transitory computer readable medium, the all-wheel drive system, comprising a center differential configured to distribute torque input thereto between front wheels and rear wheels of a vehicle, a limited-slip differential clutch configured to limit a differential operation of the center differential in accordance with an engaging pressure and change a front-rear torque distribution ratio between the front wheels and the rear wheels, a front-wheel torque transmission system configured to transmit torque between the center differential and the front wheels, and a rear-wheel torque transmission system configured to transmit torque between the center differential and the rear wheels, the program executable by processor and configured to: adjust the engaging pressure of the limited-slip differential clutch based on a driving state of the vehicle; and setting a reduction ratio of the front-wheel torque transmission system and a reduction ratio of the rear-wheel torque transmission system to be

different from each other, and wherein the center differential is configured such that the front-rear torque distribution ratio is initially unequal and is changeable.

14. The program for the all-wheel drive system according to claim 13, wherein the front-rear torque distribution ratio of the center differential is initially front biased, and wherein a gear ratio of a transfer gearset, a gear ratio of a front differential, and/or a gear ratio of a rear differential are set so that the reduction ratio of the front-wheel torque transmission system is less than the reduction ratio of the rear-wheel torque transmission system.

15. The program for the all-wheel drive system according to claim 14, further comprising adjusting the engaging pressure of the limited-slip differential clutch in a case where the driving state of the vehicle is accelerated cornering or accelerated climbing, and disengages the limited-slip differential clutch in a case where the driving state of the vehicle is other than the accelerated cornering or the accelerated climbing.

16. The program for the all-wheel drive system according to claim 14, further comprising increasing the engaging pressure of the limited-slip differential clutch in a case where the torque input to the center differential is greater than or equal to a predetermined torque and a steering angle of a steering wheel is greater than or equal to a predetermined angle, or in a case where the torque input to the center differential is greater than or equal to a predetermined torque and a pitch angle of the vehicle is greater than or equal to a predetermined angle.

17. The program for the all-wheel drive system according to claim 14, further comprising disengaging the limited-slip differential clutch in a case where an oil temperature of the limited-slip differential clutch is higher than or equal to a predetermined temperature.

18. The program for the all-wheel drive system according to claim 13, further comprising adjusting the engaging pressure of the limited-slip differential clutch in a case where the driving state of the vehicle is accelerated cornering or accelerated climbing, and disengages the limited-slip differential clutch in a case where the driving state of the vehicle is other than the accelerated cornering or the accelerated climbing.

19. The program for the all-wheel drive system according to claim 13, further comprising increasing the engaging pressure of the limited-slip differential clutch in a case where the torque input to the center differential is greater than or equal to a predetermined torque and a steering angle of a steering wheel is greater than or equal to a predetermined angle, or in a case where the torque input to the center differential is greater than or equal to a predetermined torque and a pitch angle of the vehicle is greater than or equal to a predetermined angle.

20. The program for the all-wheel drive system according to claim 13, further comprising disengaging the limited-slip differential clutch in a case where an oil temperature of the limited-slip differential clutch is higher than or equal to a predetermined temperature.
